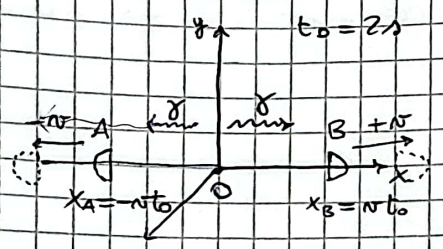
CORREZIONE esame SCRITTO 05/07/2023

1) $\Delta t' = \gamma\left(\Delta t + \frac{v}{c^2}\Delta x\right)$

(poiché si muoveva verso Sn



$\Delta t = 0$

$ct = x_A + vt \longrightarrow t = \frac{x_A}{c-v} = \frac{vt_0}{c-v}$

$x_{OA} = \frac{cvt_0}{c-v}$

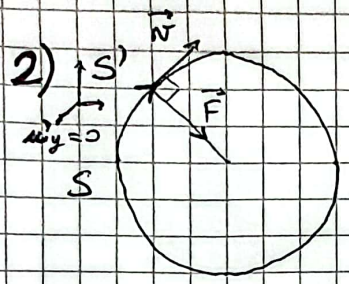
$\Delta x = \frac{2cvt_0}{c-v} = 18 \cdot 10^8\ m$

$\longrightarrow \Delta t' = 1{,}25\left(\frac{0{,}6c}{c^2} \cdot 18 \cdot 10^8\right) = 4{,}5\ s$

- leggere bene
- chiedo in S'
- cosa conosco in S

utilizzando le trasformazioni di Lorentz NON si sbaglia mai

2)



$$\vec{F} = m\gamma \vec{a} = \underbrace{m\gamma \frac{v^2}{R} = qvB}$$

$$B = \frac{m\gamma v}{qR}$$

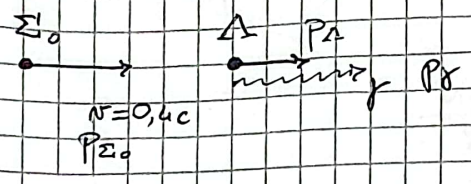
oppure $v \approx c$
oppure $p \approx E$

$$\alpha = \gamma^2 a = \gamma^2 \frac{v^2}{R}$$

$$F' = m\alpha$$

$$F' = \gamma F$$

3)



$$p_\Lambda + p_\gamma = p_{\Sigma_0} \qquad E_{\Sigma_0} = m_{\Sigma_0}\gamma c^2$$

$$E_\Lambda = \sqrt{p_\Lambda^2 c^2 + m_\Lambda^2 c^4}$$

$$E_\gamma = E_i$$

$$E_\Lambda + E_\gamma = E_{\Sigma_0}$$

$$(E_{\Sigma_0} - E_\gamma)^2 = E_\Lambda^2$$

$$m_{\Sigma_0}^2 c^4 + \cancel{p_{\Sigma_0}^2 c^2} + \cancel{p_\gamma^2 c^2} - 2E_{\Sigma_0}E_\gamma = m_\Lambda^2 c^4 + p_\Lambda^2 c^2 =$$

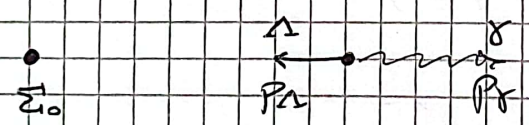
$$= m_\Lambda^2 c^4 + \cancel{p_{\Sigma_0}^2 c^2} + \cancel{p_\gamma^2 c^2} + p_\gamma^2 c^2 - 2p_{\Sigma_0}E_\gamma c$$

$$\rightarrow E_\gamma = \frac{m_{\Sigma_0}^2 c^4 - m_\Lambda^2 c^4}{2(E_{\Sigma_0} - p_{\Sigma_0}c)}$$

oppure

calcolo decadimento nel sdr del CM

S'



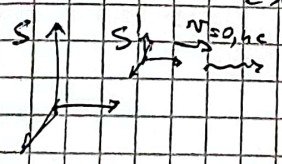
$$p_\gamma = p_\Lambda = p$$

$$E_{\Sigma_0} = E_\Lambda - E_\gamma$$

$$(E_{\Sigma_0} - E_\gamma)^2 = E_\Lambda^2$$

$$m_{\Sigma_0}^2 c^4 + p_\gamma^2 c^2 - 2m_{\Sigma_0}c^2 E_\gamma = m_\Lambda^2 c^4 + p_\Lambda^2 c^2$$

$$E_\gamma' = \frac{m_{\Sigma_0}^2 c^4 - m_\Lambda^2 c^4}{2m_{\Sigma_0}c^2} = 74,5\,MeV$$



$$E_\gamma = E_\gamma' \sqrt{\frac{1+\beta}{1-\beta}}$$

$$E = \gamma(E' + \beta p' c)$$